6.6 → Frequency-Severity Techniques

6.6.1 → Frequency-Severity Technique #1

→ Assumptions + uses

→ Frequency-Severity technique #1 applies the development technique separately to claim counts + claim severities.

→ Here are the three key assumptions for this technique:

→ 1) Claim counts + severities recorded to date will continue to develop in a similar fashion in the future.

→ 2) Claim counts have a consistent definition throughout the experience period.

→ For instance, it is not appropriate to combine claimant counts w/ occurrence counts, where all claimants under a single occurrence are recorded as one claim, unless the method of counting is consistent between the two.

→ 3) The mix of claim types is reasonably homogeneous

→ For instance, it would not be reasonable to combine the average values of slip-and-fall claims w/ those from class action lawsuits involving thousands of insured parties, as the resulting average would have little significance.

→ The second + third assumptions are common between all three of the frequency-severity techniques

→ Technique

→ Frequency-Severity #1 technique (FS #1) can be broken into four main steps:

→ 1) Project + select ultimate claim counts

→ 2) Project + select ultimate severity

→ 3) Project ultimate claims as ultimate counts times ultimate severity

→ 4) Compute unpaid claim / IBNR estimate (if needed)

→ Essentially, we're just performing the development technique on claim counts + severities separately to get ultimate claim counts + ultimate severities. Then, for each AY, we project the ultimate claims by multiplying ultimate claim counts by ultimate severity. Finally, we subtract paid claims to date to estimate unpaid claims or subtract reported claims to date to estimate IBNR (broad definition).

→ Frequency-Severity techniques

→ #1 ⇒ just develop freq + sev separately
↳ claim counts

→ #2 → — — — — — — — —, but w/ trend

→ #3 → Disposal rate method

Be sure to calculate Severity = $\frac{\text{Losses}}{\text{Claim Count}}$ + don't just use Losses (sometimes called "claims")

6.6.2 → Frequency-Severity technique #2

→ Assumptions + uses

→ Similar to Technique #1, Frequency-Severity technique #2 also relies on the development method. However, technique #2 incorporates trend analysis for exposure, frequency, + severity. Given its reliance on the development method, it shares the same three key assumptions as Technique #1:

→ 1) Claim counts + severities recorded to date will continue to develop in a similar fashion in the future.

→ 2) Claim counts have a consistent definition throughout the experience period

→ 3) The mix of claim types is reasonably homogeneous

→ This technique is especially useful when estimating the ultimate loss or IBNR for AYs that are relatively immature, especially if the CDFs for the early years are highly leveraged. Although it relies on obtaining CDFs using the development method, they are often (though not always) provided during the exam.

= lots of variation (for CDFs, it means it's large)

→ Technique

→ FS #2 technique can be broken into five steps:

1) Project ultimate claim counts
2) Compare claim counts to exposures, + select ultimate frequency
3) Project + select ultimate severity
4) Project ultimate claims
5) Compute unpaid claim / IBNR estimate (if needed)

→ Essentially, we select a frequency/severity as the average historical ultimate frequency/severity trended to the year of interest. Then, we calculate an ultimate claims estimate for that year as

$$\text{AY ult claims} = \underbrace{\text{Projected Claim Counts}}_{\text{AY Exposures} * \text{Selected Frequency}} * \text{Selected Severity}$$

→ F/S #2 technique ⇒ don't use year we are estimating losses for in our calculation
(just like w/ the EC method)

6.6.3 → Disposal rate method

→ Assumptions + uses

→ The final frequency-severity technique we will explore is often referred to as the disposal rate method. The disposal rate for a particular AY + development year is determined by dividing the cumulative closed claim counts by the ultimate closed claim counts. In other words, it is the percentage of ultimate claims that have been closed.

$$\text{AY Disposal Rate}_t = \frac{\text{AY Cumulative Closed Claim Counts}_t}{\text{AY Ultimate Closed Claim Counts}}$$

where subscript t represents the maturity (12 months, 24 months, etc.)

→ Like the other FS techniques we've discussed, the disposal rate method utilizes the development technique. Therefore, the same three key assumptions apply. In addition, the disposal rate method also makes a fourth assumption:

→ 1) Claim counts + severities recorded to date will continue to develop in a similar fashion in the future.

→ 2) Claim counts have a consistent definition throughout the experience period

→ 3) The mix of claim types is reasonably homogeneous

→ 4) There are no significant partial claim payments made.

→ To clarify, the fourth assumption essentially means that the disposal rate method assumes that exactly one payment is made on each claim. So, although the reported amount for an individual claim may develop, it is not paid until it is fully developed. Once it is fully developed, a single payment is made for the full amount of the claim, + the claim closes.

→ Technique

→ The disposal rate method can be broken into seven steps:

→ 1) Project + select ultimate claim counts by AY

→ 2) Create disposal rate triangle + select disposal rates by maturity

→ 3) Project claim counts using the selected disposal rates → $\text{open claims} * \frac{\Delta \text{DR}_{\text{remaining}}}{1 - \text{Selected DR}}$
↳ for each t_2, t_1

→ 4) Analyze + select incremental severities by maturity

→ 5) Calculate incremental severities by maturity age + AY

→ 6) Project unpaid claims by multiplying claim counts by severities

→ 7) Compute ultimate claim / IBNR estimate (if needed)

Accident Year	Incremental Severity Trended to AY2023		
	12-24 Months	24-36 Months	36-48 Months
2020	$4{,}000(1.02)^{3} = 4{,}244.83$	$6{,}000(1.02)^{3} = 6{,}367.24$	$10{,}000(1.02)^{3} = 10{,}612.08$
2021	$4{,}030(1.02)^{2} = 4{,}192.81$	$5{,}950(1.02)^{2} = 6{,}190.38$	
2022	$4{,}050(1.02) = 4{,}131$		

★ Table just has to be trended to the same year, doesn't matter which year for when we are making a selection

→ just picked latest year so don't have to de-trend anything

⇒ Then we get same answer at end b/c we then re-trend to whatever year we are estimating for (if needed)

⇒ We trend w/ ( )^3 each time b/c we are trending the incremental severities for each maturity to AY 2023, not CY 2023. So the trend / inflation levels should also correspond to the AY, not the CY.

→ note that the 12 to 24-month severity for AY 2023 would correspond to claims that occur during 2023 but are made from 1/1/2024 through 12/31/2024.

→ Frequency-Severity technique #3 (Disposal rate method)

→ Big picture steps → 1) Calc incremental paid severities
2) Trend to latest AY + select
3) Calc disposal rates + select
4) Calc est incremental closed claims, CHECK: sum (closed) = ultimate
Portion out remaining open claims based on disposal rates
5) Est unpaid claims as selected sev * selected closed claims

6.7 → Case Outstanding techniques

→ 6.7.1 → Case outstanding Technique #1

→ Assumptions + uses

→ Like many of the other reserving methods we've discussed, this technique is essentially a variation of the development method. Instead of developing claims, we develop case outstanding + use the developed case outstanding to project claims. Since this technique relies on the development method, it has the same three key assumptions w/ the exception that the second assumption pertains to case outstanding instead of observed claims:

→ 1) Claims recorded to date will continue to develop in a similar fashion in the future

→ 2) Case outstanding gives us relevant information on claims that have yet to be observed

→ 3) Throughout the policy period:

→ The mix of claim types is stable

→ Policy limits (if any) are stable

→ Reinsurance retention limits (if any) are stable

→ There is consistent claims processing (claims settlement rates + case outstanding adequacy)

→ Another disadvantage is that the ratios used in Technique #1 are somewhat difficult to select + interpret as compared to development factors. Furthermore, large claims during the experience period can distort case outstanding which would, in turn, impact the unpaid claims estimates.

→ On the bright side, case outstanding technique #1 is great when most or all claims are reported during the first development period. It pairs very well w/ claims-made coverages or report-year analyses b/c there is no pure IBNR.

→ Technique

→ The remaining-in-case ratio for a given age + maturity is the case outstanding at that maturity divided by the case outstanding for the previous maturity. It is essentially the percent of the previous period's case outstanding that is "remaining".

$$\text{Remaining-in-Case ratio} = \text{Ratio of case outstanding to previous case outstanding} = \frac{\text{Current Case Outstanding}}{\text{Prior Case Outstanding}}$$

→ The paid-on-case ratio for a given age + maturity is the incremental amount paid for that period divided by the case outstanding at the end of the previous period. This is essentially the percentage of the previous period's case outstanding that are paid this period.

$$\text{Paid-on-case ratio} = \text{Ratio of incremental paid claims to previous case outstanding} = \frac{\text{Incremental paid claims}}{\text{Prior Case outstanding}}$$

→ Assuming we start w/ case outstanding + incremental paid claims triangles, this method can be summarized in four steps:

→ 1) Calculate + select the remaining-in-case ratios for each maturity

→ 2) Project case outstanding using the selected remaining-in-case ratios

$$\text{Projected case outstanding} = \text{Selected Remaining-in-case ratio} * \text{Prior Case outstanding}$$

(This is analogous to projecting claims under the development method.)

→ 3) Calculate + select the paid-on-case ratios for each maturity

→ 4) Project incremental payments using the selected paid-on-case ratios

$$\text{Projected Incremental paid claims} = \text{Selected Paid-on-case ratio} * \text{Prior case outstanding}$$

6.7.2 → Case outstanding Technique #2

→ Assumptions + uses

→ In the event that an insurer lacks historical claim data, case outstanding Technique #2 is a viable option. It allows us to project future unpaid claims by combining case outstanding using industry-based development factors. Thus, the only information this technique requires is the insurer's current case outstanding.

→ Technique #2 has the same three key assumptions as Technique #1, with the exception that future development is based on industry-reported/paid CDFs.

→ 1) Claims recorded to date will develop in a similar fashion in the future to the industry benchmark

→ 2) Case outstanding gives us relevant information on claims that have yet to be observed

→ Throughout the policy period:

→ The mix of claims is stable

→ Policy limits (if any) are stable

→ Reinsurance retention limits (if any) are stable

→ There is consistent claims processing (claim settlement rates + case outstanding adequacy).

→ Like Technique #1, Technique #2 is not commonly used. However, there are instances where case outstanding could be the only information available to the actuary. This is most common for self-insurers or in the aftermath of mergers or consolidations w/ companies that are self-insured, especially for older years.

→ There are several disadvantages to this technique, however. It relies on the insurer's ability to obtain industry CDFs, which may not always be feasible. Additionally, it assumes that the obtained industry CDFs are representative of the future claim development w/in the company. Lastly, any anomalies or large losses in the case outstanding can distort the results of this approach

→ Technique

→ The main idea of case outstanding Technique #2 is to combine industry-based paid + reported CDFs to create a singular case outstanding development factor.

$$\text{Case outstanding Development factor} = 1 + \frac{(\text{Reported CDF} - 1)(\text{Paid CDF})}{\text{Paid CDF} - \text{Reported CDF}}$$ ⇒ just memorize this

→ The factor is calculated by AY. We then multiply it by the case outstanding for the corresponding AY to project unpaid claims.

7.1 → Berquist-Sherman techniques

7.1.1 → Berquist-Sherman Paid Claim development adjustment

→ Overview → As we discussed in Section 6.2, a change in the rate at which claims are settled can distort the age-to-age factors when using the development method. To be specific, an increase in the settlement rate can result in overestimated development factors + a decrease in the settlement rate can result in underestimated development factors. This would lead to overestimated or underestimated ultimate claims estimates, respectively.

→ To account for this, we can apply the Berquist-Sherman paid claims development adjustment, sometimes called the Berquist-Sherman claims settlement rate adjustment. This technique involves adjusting paid claims by interpolating between the historical disposal rates + the most recent disposal rates at each maturity for the same for each AY. After this, the development method can be performed as usual.

→ This technique's main assumption is that increases in the percentage of closed claim counts relative to ultimate claim counts are associated w/ increases in the percentage of total claims paid. In other words, the disposal rates are assumed to be roughly proportional to the total percentage of ultimate claims paid for each maturity.

→ Mechanics of the Berquist-Sherman paid claim development adjustment

→ In order to determine whether or not the B-S adjustment is needed, we should have evidence that the settlement rate is actually changing. We can do this by looking at a triangle of the disposal rates. If the disposal rates at each maturity are not consistent between AYs, an adjustment may be appropriate

→ 1) Analyze the historical disposal rates. If disposal rates have changed significantly over the years, a B-S paid claims development adjustment is warranted.

→ 2) (Assuming adjustment is appropriate) Restate the cumulative paid claims.

→ a) Select the disposal rates by maturity

★ → b) Interpolate between paid claims of consecutive maturities

→ If historical disposal rate < selected disposal rate, then adjust paid claims upward w/ equation (1)

→ If historical disposal rate > selected disposal rate, then adjust paid claims downward w/ equation (2)

→ 3) Perform the paid development method

★ linear interpolation

★ Excel
= Forecast (x, [y₀:y₁], [x₀:x₁])
↳ output is adjusted P_c
⇒ works regardless of adjusting upward or downward (i.e. no need for two cases of the formula)
⇒ NEED column of zeros at 0 months for it to work for all cells

→ or exponential interpolation

⇒ At each maturity t, $b = \frac{\ln(\text{claims at } t) - \ln(\text{claims at } t-1)}{[\text{Count at } t] - [\text{Count at } t-1]} = \frac{\Delta y}{\Delta x}$ slope

$$\Rightarrow \ln(y) = \ln(a e^{bx}) = \ln(a) + bx$$

$$e^{\ln(y) - bx} = e^{\ln(a)}$$

$$\Rightarrow a = \frac{y}{e^{bx}} = \frac{\text{claims at } t}{\exp[b \times \text{count at } t]}$$

recalculate for each AY + maturity combo

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1															
2		Accident	Closed Claim Counts				Projected Ultimate			Disposal rates					
3		Year	0 Months	12 Months	24 Months	36 Months	Claim Counts		Accident Year	0 Months	12 Months	24 Months	36 Months		
4		2009	0	425	700	850	850		2009	0.00	0.500	0.824	1.000		
5		2010	0	400	693		887		2010	0.00	0.451	0.781	=E5/$G5		
6		2011	0	373			901		2011	0.00	0.414				
7															
8									-> Decreasing claim settlement rates; when interpolating, always use downward adjustment						
9		Accident	Cumulative Paid Claims ($000s)												
10		Year	0 Months	12 Months	24 Months	36 Months									
11		2009	0	4,200	8,580	10,000			Age	0 Months	12 Months	24 Months	36 Months	=M3	
12		2010	0	3,950	7,030				Selected DR	0.000	0.414	0.781	1.000	=MIN(M4:M6)	
13		2011	0	3,900											
14															
15									Accident Year	Adjusted cumulative Paid Claims ($000s)					
16										12 Months	24 Months	36 Months			
17									2009	3,477	8,008	10,000	=FORECAST(M12,E11:F11,$L4:$M4)		
18									2010	3,626	7,030	=FORECAST(L12,D12:E12,$K5:$L5)			
19									2011	3,900	=FORECAST(K12,C13:D13,$J6:$K6)				
20															
21															
22									Accident Year	Development factors					
23										12-24 Months	24-36 Months				
24									2009	2.30	1.25	=L17/K17			
25									2010	1.94					
26															
27															
28									Age-to-age	12-24 Months	24-36 Months	=K23			
29									Selected factor	2.12	1.25	=AVERAGE(K24:K25)			
30															
31															
32									AY 2011 est ult claims =	$10,328,332.28	=D13*PRODUCT(J29:K29)*1000				

7.1.2 → Berquist-Sherman Reported Claim development adjustment

→ Overview → Similar to the way a change in the claims settlement rate can impact paid claims data, a change in the case outstanding adequacy can impact reported claims data. If a company begins increasing case outstanding for each reported claim, the development method will interpret the increase in reported claims as an increase in severity. If the claim severity isn't actually increasing, development factors + ultimate claims estimates will be overstated. Likewise, a decrease in case outstanding adequacy would cause development factors + ultimate claims estimates to be understated.

→ We can again turn to Berquist + Sherman for a solution. The Berquist-Sherman reported claims development adjustment, or the Berquist-Sherman case outstanding adequacy adjustment, involves restating reported claims data at a common level of case outstanding adequacy. After this, the reported development method can be performed using the adjusted reported data.

→ The reported B-S technique assumes that annual changes in the average case outstanding at each maturity are due to changes in case outstanding adequacy or trends in claim severity. In practice, there are other factors that could impact the average case outstanding such as increasing the amount of emphasis placed on settling large/small claims earlier on.

→ Mechanics of the Berquist-Sherman case outstanding adjustment

→ As w/ the paid method, it is important to be able to determine when the reported B-S adjustment is necessary. An adjustment is appropriate when there have been changes in case outstanding adequacy. We can check this by studying a triangle of the average case outstanding.

$$\underbrace{\text{Avg Case OS}}_{\text{"outstanding"}} = \frac{\text{Case OS}}{\text{Open Claim Counts}} = \frac{\text{Reported Claims} - \text{Paid Claims}}{\text{Reported Counts} - \text{Closed Counts}}$$

→ In general, the average case outstanding for each maturity should trend at the same rate as severity. For example, if the 2020 average case outstanding at 12 months is \$200, + claim severity trends at 5% annually, then we would expect the 2021 average case outstanding at 12 months to be \$200 (1.05) = \$210. If it is not, then the reported claims data might be distorted by a change in case outstanding adequacy.

→ Here is a summary of the B-S reported claims development adjustment:

→ 1) Evaluate the data

→ a) Analyze the % change in claim severities

→ b) Analyze the % change in average case outstanding

→ c) If the changes deviate significantly, a B-S reported claims development adjustment is warranted

→ 2) (Assuming adjustment is appropriate) Restate the cumulative reported claims

→ a) Select a severity trend (if a severity trend is given by the question, we can use it directly).

→ b) Adjust the average case outstanding by trending the most recent data backward.

→ c) Restate the cumulative reported claims using the adjusted average case outstanding

→ 3) Perform the reported development method

$$\text{Adj Reported claims} = \text{Adj Avg Case OS} * \underbrace{\left(\text{Reported Counts} - \text{Adj Closed Counts}\right)}_{\text{Adj open claim counts}} + \text{Adj Paid claims}$$

Accident Year	Adjusted Average Case Outstanding ($000)			
	12 Months	24 Months	36 Months	48 Months
2011	$5.922(1.05)^{-3} = 5.116$	$5.060(1.05)^{-2} = 4.590$	$4.714(1.05)^{-1} = 4.490$	0
2012	$5.922(1.05)^{-2} = 5.372$	$5.060(1.05)^{-1} = 4.819$	4.714	
2013	$5.922(1.05)^{-1} = 5.640$	5.060		
2014	5.922			

→ Berquist-Sherman method ⇒ adjusts losses for change in case reserve adequacy

→ 0) To determine if necessary, check if avg case outstanding trends the same as severity
↳ = $\frac{\text{case outstanding}}{\text{open claims}}$ → = rept claims − paid claims

→ 1) Trend the most recent avg case outstanding backwards in time
(using the severity trend, b/c they should align)

→ 2) Calculate adjusted cumulative rept claims = adj avg case outstanding * open claim count + paid claims

→ 3) Then do dev method like usual

⇒ results, if strengthening case reserves, using regular dev method will overstate results
b/c LDFs calculated on the unadjusted data would be applied to higher rept loss (from the strengthened reserves) will overstate the ultimate loss

⇒ Shortcoming is it is based highly on the selected severity
↳ think about any assumptions made in the calculations